MATH 189 HW 2

Dani Nguyen

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Question 1

a. Since IQ and GPA is fixed, we will denote those values as a constant C

$$\begin{aligned} Y &= C + 35X_3 - 10X_5 \\ &= C + \mathbb{I}\{\text{has college}\}(35 - 10X_5) \end{aligned}$$

When fixing IQ and GPA, the model predicts that college graduates will earn less if their GPA is higher than 3.5. Therefore, option iii. is correct.

b. For a college graduate with IQ 110 and GPA 4.0,

$$\begin{aligned} Y &= 50 + 20(4.0) + 0.07(110) + 35(1) + 0.01(4.0 \cdot 110) - 10(4.0 \cdot 1) \\ &= 137.1 \text{ thousand dollars} \end{aligned}$$

c. False, because GPA and IQ are on average very large numbers, especially when multiplied, that even a low coefficient can lead to large effects in salary. This does not mean that there is little evidence of an interaction effect, one would need to look at the p-values of the interaction term to know whether there exists an interaction effect.

Question 2

$$
\begin{aligned}
\hat{y}_i &= x_i\hat{\beta} \\
&= x_i \frac{\sum_{i=1}^{n} x_i y_i}{\sum_{i'=1}^{n} x_{i'}^2} \\
&= \frac{\sum_{i=1}^{n} x_i^2 y_i}{\sum_{i'=1}^{n} x_{i'}^2} \\
&= \sum_{i=1}^{n} \frac{x_i^2}{\sum_{i'=1}^{n} x_{i'}^2} y_i
\end{aligned}
$$

Let $a_i = \frac{x_i^2}{\sum_{i'=1}^{n} x_{i'}^2}$

$$
\hat{y}_i = \sum_{i=1}^{n} a_i y_i
$$

Question 3

Recall that $\beta_0 = \bar{y} + \beta_1\bar{x}$. This can be rearranged to $\bar{y} = \beta_0 - \beta_1\bar{x}$. This means that the equation for simple linear regression is centered at and will always pass through the point $(\bar{x}, \bar{y})$.

Question 4

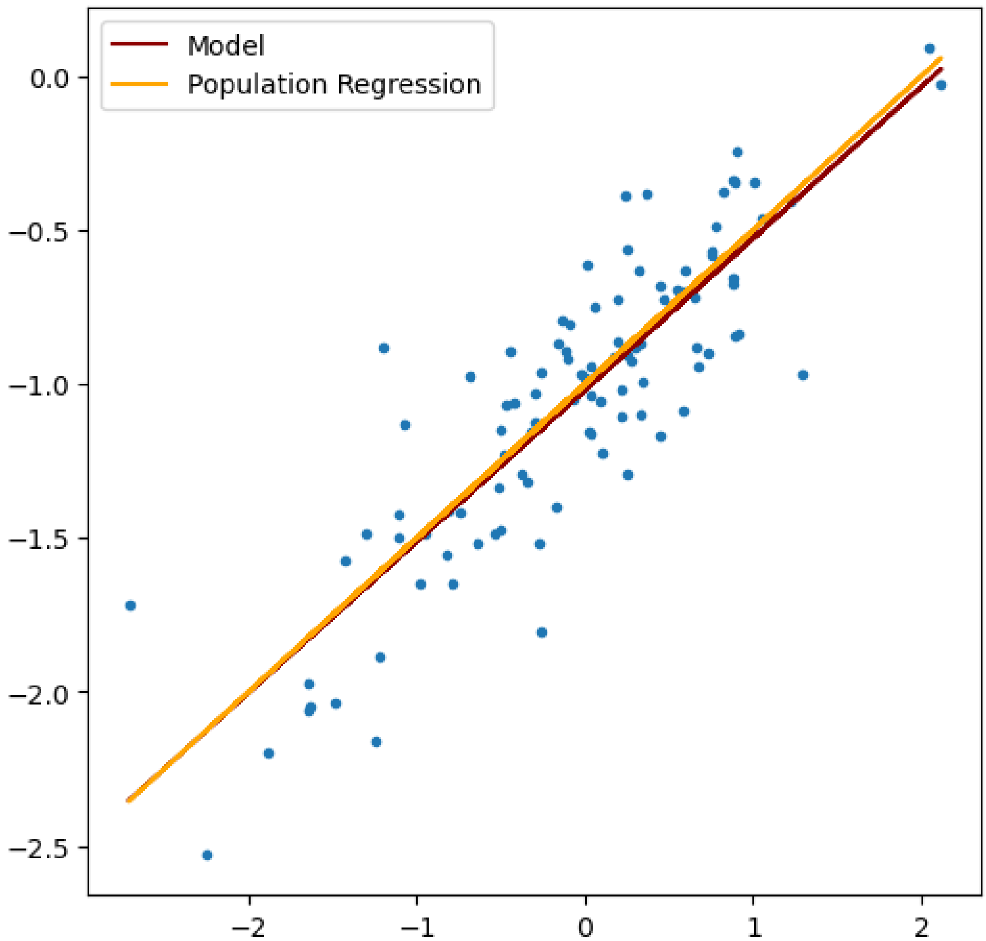
$$
\begin{aligned}
R^2 &= \frac{TSS - RSS}{TSS} \\
&= \frac{\sum_{i=1}^{n}(y_i)^2 - \sum_{i=1}^{n}(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2}{\sum_{i=1}^{n}(y_i)^2} \\
&= \frac{\sum_{i=1}^{n}(y_i)^2 - \sum_{i=1}^{n}(y_i)^2 + \frac{\left[\sum_{i=1}^{n}(x_i-\bar{x})(y_i-\bar{y})\right]^2}{\sum_{i=1}^{n}(x_i-\bar{x})^2}}{\sum_{i=1}^{n}(y_i)^2} \\
&= \frac{\sum_{i=1}^{n}(x_i y_i)^2}{\sum_{i=1}^{n}(x_i)^2 \sum_{i=1}^{n}(y_i)^2} \\
R &= \frac{\sum_{i=1}^{n}(x_i y_i)}{\sum_{i=1}^{n}(x_i) \sum_{i=1}^{n}(y_i)} = \frac{Cov(X_i Y_i)}{Var(X_i)^{\frac{1}{2}} Var(Y_i)^{\frac{1}{2}}}
\end{aligned}
$$

Question 5

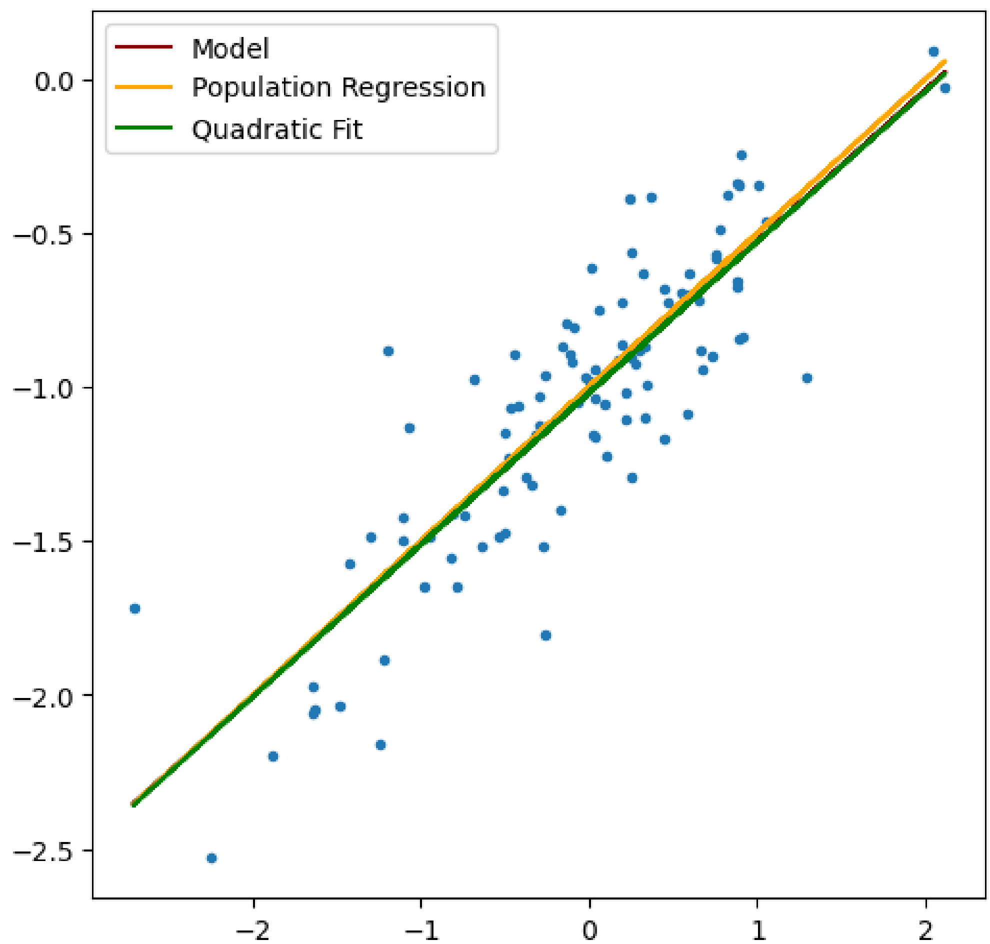
c. Length of the vector y is 11.62. In this linear model β_0 is -1 and β_1 is 0.5.

d. The distribution is linear, positive, and moderately spread out.

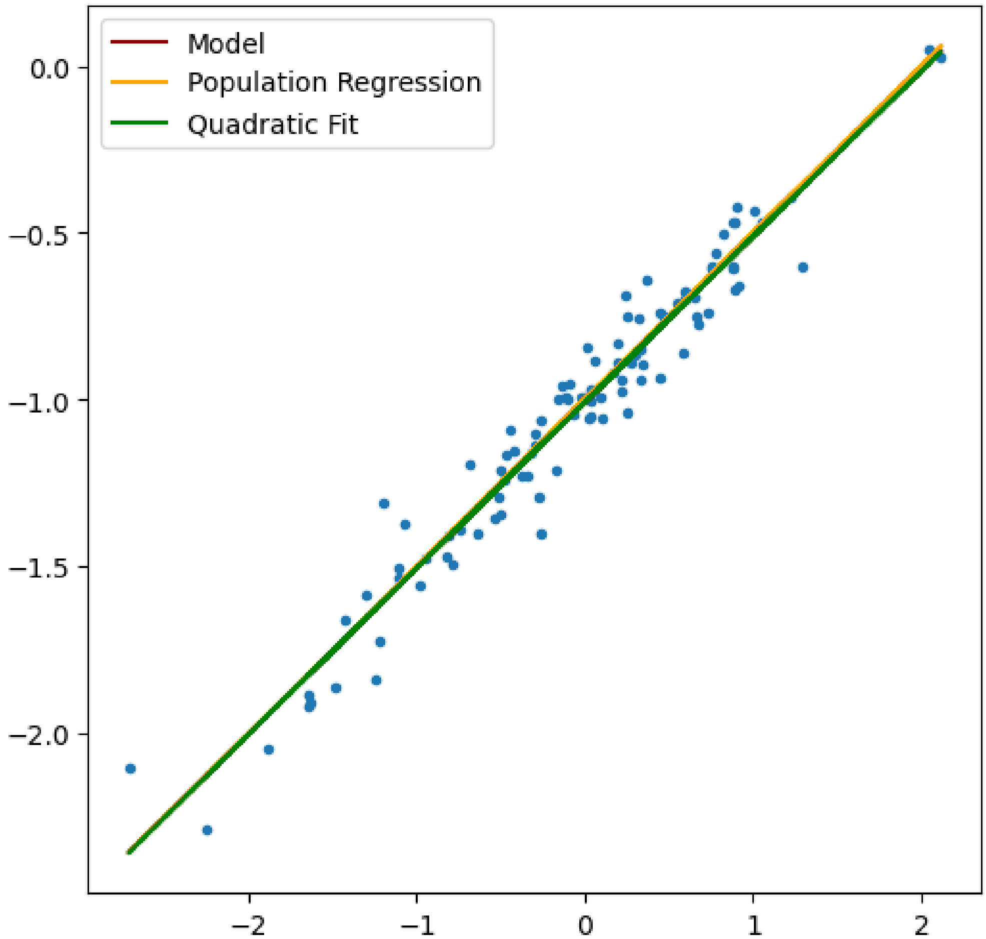
e. The model is linear and has positive slope, $\hat{\beta}_1$ is 0.49214551 and $\hat{\beta}_0$ is -1.01900639 which is fairly close to our β_1 of 0.5 and β_0 of -1

f. 

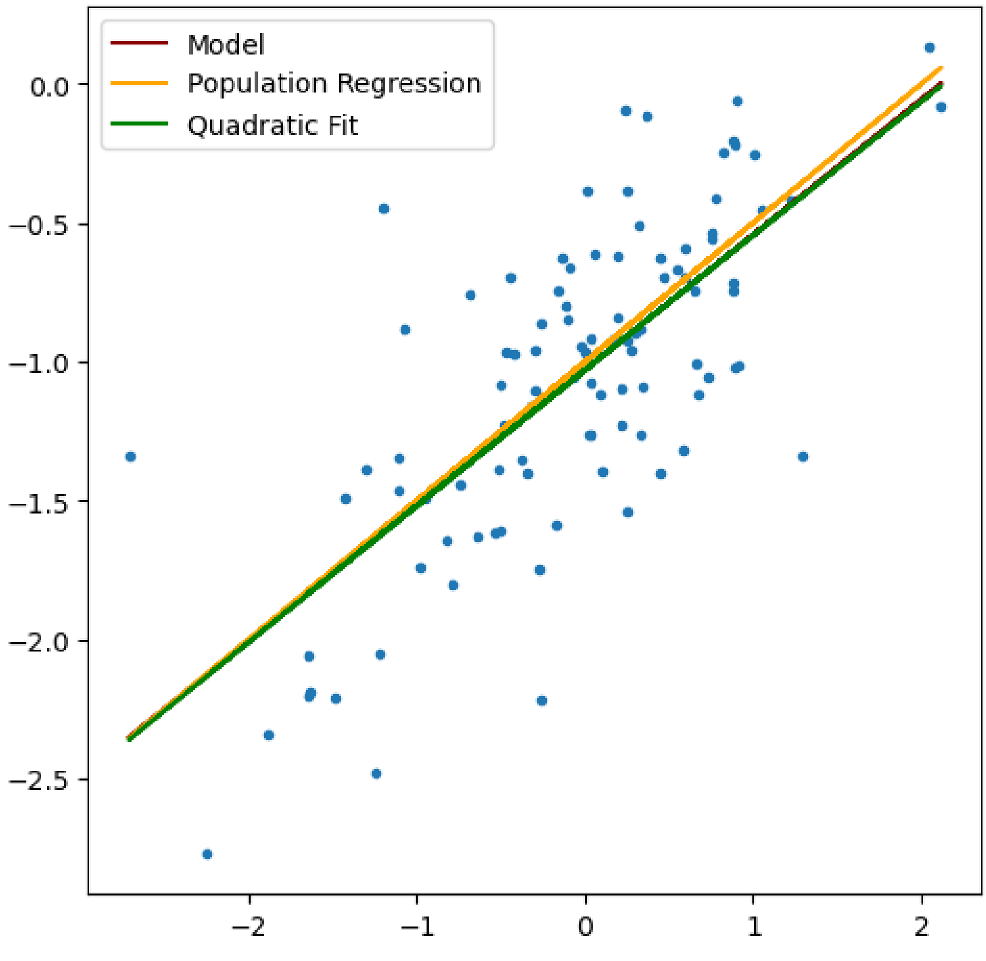
g. The quadratic line of best fit seems to fit just as well as the linear model, although there is no evidence that it improves the fit.



h. The points are much closer to the least squares regression line, and the differences between the model and the population regression decreases



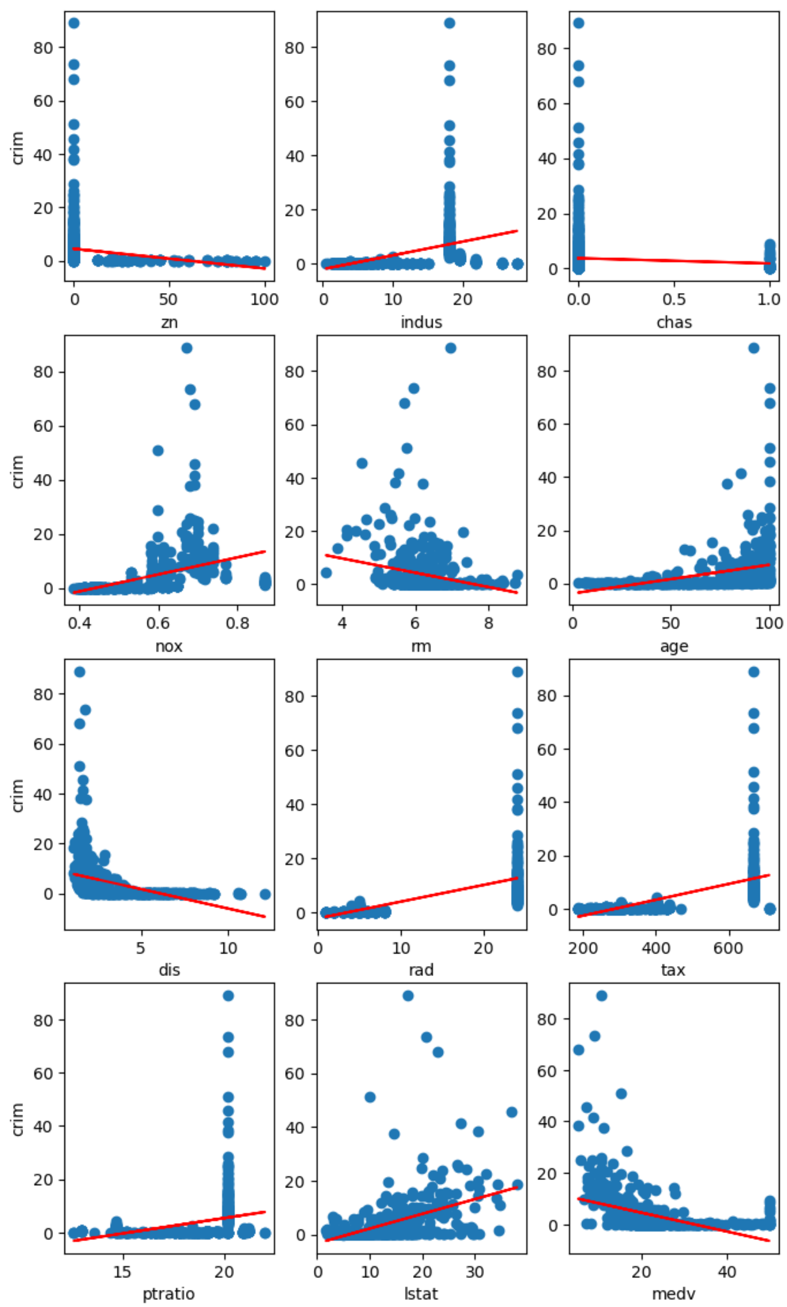
i. The points are more spread out along the least squares regression line, and the differences between the model and the population regression increases



j. For the noisy dataset, the confidence interval for β_0 is $[-1.03 \pm 0.0798]$, β_1 is $[0.4874 \pm 0.0933]$.
For the original dataset the confidence interval for β_0 is $[-1.02 \pm 0.0499]$, β_1 is $[0.4921 \pm 0.0584]$.
For the less noisy dataset the confidence interval for β_0 is $[-1.007 \pm 0.0200]$, β_1 is $[0.4969 \pm 0.0233]$.

Question 6

None of the distributions seem to fit a linear regression model, but there does seem to be significant association with some of the data. For example, lstat has a 0.4556 correlation coefficient with crim, nox is 0.421,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